PGCP-AC AUGUST 26
Assignment No- 2

1. Write a SQL query to create Employee & Department table with specific columns and constraints.(**Note**- Create Table, insert data as per your Choice)

ANS ->

CREATE TABLE Department (DeptID INT PRIMARY KEY, DeptName VARCHAR(50) UNIQUE NOT NULL);

CREATE TABLE Employee (EmpID INT PRIMARY KEY, EmpName VARCHAR(50) NOT NULL, Salary DECIMAL(10,2), JoiningDate DATE, DeptID INT, City VARCHAR(50), JobTitle VARCHAR(50), FOREIGN KEY (DeptID) REFERENCES Department(DeptID));

2. Write a SQL query to add any new column to an existing table.

ANS -> ALTER TABLE employee ADD COLUMN Email VARCHAR(100);

3. Write a SQL query to insert multiple records (At least 5) into a table in a single operation.

ANS ->

INSERT INTO Employee (EmpID, EmpName, Salary, JoiningDate, DeptID, City, JobTitle) VALUES

(106, 'Neha', 52000, '2024-04-10', 10, 'Pune', 'Developer'),

(107, 'Karan', 48000, '2023-07-15', 20, 'Mumbai', 'HR Executive'),

(108, 'Meera', 65000, '2022-11-20', 30, 'Mumbai', 'Salesperson'),

(109, 'Rohan', 58000, '2024-02-01', 40, 'Nagpur', 'Accountant'),

(110, 'Pooja', 43000, '2025-01-12', 20, 'Pune', 'HR Executive')

;

4. Write a SQL query to delete all records from a table.

ANS -> DELETE FROM Employee;

5. Write a SQL query to update salary records from a table where id = 5.

ANS -> UPDATE Emp SET Salary = 55000 WHERE id = 5;

6. Write a query to grant SELECT privilege on the Emp table to user readonly_user.

ANS -> GRANT SELECT ON Emp TO readonly_user;

7. Write a query to revoke INSERT privilege on the Emp table from user temp_user.

ANS -> REVOKE INSERT ON Emp FROM temp_user;

8. Write a query to insert a new employee into the Emp table and commit the transaction.

ANS -> INSERT INTO Emp (id, EmpName, Salary) VALUES (101, 'Amit', 60000);

COMMIT;

9. Write a query to update the Salary of an employee with EmpID = 101 and commit the transaction.

ANS -> UPDATE Emp SET Salary = 65000 WHERE EmpID = 101;

COMMIT;

10. Write a query to set a SAVEPOINT before updating the DeptID of an employee.

ANS -> SAVEPOINT sp1;

UPDATE Emp SET DeptID = 20 WHERE EmpID = 101;

11. Write a query to rollback to the previously created SAVEPOINT.

ANS -> ROLLBACK TO SAVEPOINT sp1;

12. Write a query to truncate all data from the Emp table.

ANS -> TRUNCATE TABLE Employee;

13. Write a query to list all employees whose salary is greater than 50,000.

ANS -> SELECT * FROM Emp WHERE Salary >= 50000;

14. Write a query to increase the salary of all employees in DeptID = 10 by 10%.

ANS -> UPDATE Employee SET Salary = Salary * 1.05 WHERE DeptID = 20;

15. Write a query to count the total number of employees in each department.

ANS -> SELECT DeptID, COUNT(*) AS TotalEmployees FROM Employee GROUP BY DeptID;

16. Write a query to list departments that have no employees. & create a view for those salespeople who belong to the city of Mumbai.

ANS -> SELECT d.DeptID, d.DeptName FROM Department d LEFT JOIN Employee e ON d.DeptID = e.DeptID WHERE e.EmpID IS NULL;

17. Write a query to display the highest salary among all employees.

ANS -> SELECT MAX(Salary) AS HighestSalary FROM Employee;

18. Write a query to display the 2nd highest salary among all employees.

ANS -> SELECT MAX(Salary) AS SecondHighestSalary FROM Emp WHERE Salary < (SELECT MAX(Salary) FROM Emp);

19. Write a query to show all employees ordered by JoiningDate in descending order.

ANS -> SELECT * FROM Emp ORDER BY JoiningDate DESC;

20. Write a query to rename the DeptName column to DepartmentName.

ANS -> ALTER TABLE Department RENAME COLUMN DeptName TO DepartmentName;